Tund Theerawit

Madison, WI | Theerawit@wisc.edu | (773) 580-1515
linkedin.com/in/tund-theerawit | github.com/TundTT
https://tund-personal-website.vercel.app/

EDUCATION

University of Wisconsin–Madison Expected May 2027
B.S. Mechanical Engineering; Certificate in Data Science | GPA: 3.86/4.00

ROBOTICS EXPERIENCE

BadgerBots | UW–Madison RoboCup Team Fall 2025–Present
Low-Level Control Co-Lead (current)

- Restored a malfunctioning Booster K1 humanoid to competition-ready condition in 5 hours between match days at RoboCup; learned from Booster engineers, diagnosed and replaced a shorted motor-controller board, reflashed firmware, and validated operation.
- Adapted robot software on-site at RoboCup 2026: deployed policies, worked on whistle detection, and debugged live competition issues involving unfamiliar Wi-Fi networks and borrowed robots with different hardware.
- Co-lead a 5-person motion-control team within a ~50-member RoboCup team, coordinating policy inputs, outputs, and behavior selection across infrastructure, localization, and behavior teams so walking, kicking, and recovery work as one system.
- Competed at RoboCup in South Korea and the World Humanoid Robot Games in Beijing, where our team placed 4th out of 14 teams in robot soccer.

LeggedAI Lab | UW–Madison May 2025–Present
Undergraduate Robotics Researcher | QuadMorph

- Created end effectors that switch between wheels and legs in a heating-and-reshaping cycle; integrated hardware and learned policies on a Pupper quadruped to demonstrate rolling, walking, and transitions.
- Enabled wheel–leg transitions without extra actuators: shape-memory polymer end effectors soften with heat and reshape under robot weight, then recover their wheel shape when reheated off the ground.
- Enabled continuous wheel rotation while avoiding electrical shorts during reshaping by integrating nichrome heaters, slip rings, and wire routing around motors without through-holes.
- Used simulation-guided co-design to size 3D-printed spacers, preventing end-effector collisions with motor mounts and optimizing stance width for stability while limiting torque from offset wheels and motors.
- Demonstrated full-stack problem solving using IMU and joint-position data to detect leg-length variation and guide policy training, linking hardware diagnosis with simulation.

Research: QuadMorph paper submitted to the ICRA robotics conference; under review.

ADDITIONAL EXPERIENCE

CATS Lab, UW–Madison | Summer Undergraduate Researcher Jul–Aug 2024

AI Group, Bangkok | Project Assistant Intern Jul–Aug 2023

TECHNICAL SKILLS

Hardware and robotics: SolidWorks, Onshape, FDM and DLP resin 3D printing, mechatronics, motor-controller debugging, firmware flashing, PCB soldering, ROS 2

Programming and tools: Python, C++, Git, Linux/Bash, OpenCV, PLC/ladder logic

Simulation and learning: MuJoCo/MJX, Isaac Lab, JAX, PyTorch, PPO, domain randomization, sim-to-real transfer